

```
!unrar x /content/autonomous_driving.rar /content/
```

UNRAR 6.11 beta 1 freeware Copyright (c) 1993-2022 Alexander Roshal

Extracting from /content/autonomous_driving.rar

```
Creating      /content/autonomous_driving                OK
Extracting    /content/autonomous_driving/config.py      OK
Extracting    /content/autonomous_driving/data_generator.py OK
Extracting    /content/autonomous_driving/evaluate.py    OK
Extracting    /content/autonomous_driving/main.py        OK
Creating      /content/autonomous_driving/models          OK
Extracting    /content/autonomous_driving/models/pilotnet.py OK
Extracting    /content/autonomous_driving/models/__init__.py OK
Extracting    /content/autonomous_driving/README.md      OK
Extracting    /content/autonomous_driving/requirements.txt OK
Extracting    /content/autonomous_driving/simulator.py   OK
Extracting    /content/autonomous_driving/train.py       OK
Extracting    /content/autonomous_driving/项目文档.md    OK
Creating      /content/autonomous_driving/data            OK
Creating      /content/autonomous_driving/output          OK
Creating      /content/autonomous_driving/runs            OK
All OK
```

解压完成后，您可以使用 `(!ls /content/)` 命令查看解压后的文件。

```
!pip install -q opencv-python matplotlib tensorflow
```

```
# 验证依赖
import tensorflow as tf
import cv2
import matplotlib
print("✅ 所有依赖安装成功")
print(f"TensorFlow: {tf.__version__}")
print(f"OpenCV: {cv2.__version__}")
```

```
✅ 所有依赖安装成功
TensorFlow: 2.20.0
OpenCV: 4.13.0
```

开始借助 AI 编写或生成代码。

```
%cd /content/autonomous_driving
!python main.py generate
```

```
/content/autonomous_driving
=====
生成训练数据...
样本数: 30000
生成进度: 3000/30000
生成进度: 6000/30000
生成进度: 9000/30000
生成进度: 12000/30000
生成进度: 15000/30000
生成进度: 18000/30000
生成进度: 21000/30000
生成进度: 24000/30000
生成进度: 27000/30000
生成进度: 30000/30000
已保存 30000 个样本到 /content/autonomous_driving/data/train

生成验证数据...
样本数: 5000
生成进度: 500/5000
生成进度: 1000/5000
生成进度: 1500/5000
生成进度: 2000/5000
生成进度: 2500/5000
生成进度: 3000/5000
生成进度: 3500/5000
生成进度: 4000/5000
生成进度: 4500/5000
生成进度: 5000/5000
已保存 5000 个样本到 /content/autonomous_driving/data/val

生成测试数据...
样本数: 5000
生成进度: 500/5000
生成进度: 1000/5000
生成进度: 1500/5000
生成进度: 2000/5000
```

生成进度: 2500/5000
生成进度: 3000/5000
生成进度: 3500/5000
生成进度: 4000/5000
生成进度: 4500/5000
生成进度: 5000/5000
已保存 5000 个样本到 /content/autonomous_driving/data/test

全部数据生成完成!
训练集: /content/autonomous_driving/data/train
验证集: /content/autonomous_driving/data/val
测试集: /content/autonomous_driving/data/test

```
%cd /content/autonomous_driving
!python main.py train --model smallnet --max_samples 8000
```

/content/autonomous_driving/simulator.py:227: UserWarning: Glyph 66716 (\N{CJK UNIFIED IDEOGRAPH-1260710}) missing from font(s) DejaVu Sa
plt.savefig(save_path, dpi=150, bbox_inches="tight")
/content/autonomous_driving/simulator.py:227: UserWarning: Glyph 36742 (\N{CJK UNIFIED IDEOGRAPH-8F86}) missing from font(s) DejaVu Sa
plt.savefig(save_path, dpi=150, bbox_inches="tight")
/content/autonomous_driving/simulator.py:227: UserWarning: Glyph 34892 (\N{CJK UNIFIED IDEOGRAPH-884C}) missing from font(s) DejaVu Sa
plt.savefig(save_path, dpi=150, bbox_inches="tight")
/content/autonomous_driving/simulator.py:227: UserWarning: Glyph 39542 (\N{CJK UNIFIED IDEOGRAPH-9A76}) missing from font(s) DejaVu Sa
plt.savefig(save_path, dpi=150, bbox_inches="tight")
/content/autonomous_driving/simulator.py:227: UserWarning: Glyph 36335 (\N{CJK UNIFIED IDEOGRAPH-8DEF}) missing from font(s) DejaVu Sa
plt.savefig(save_path, dpi=150, bbox_inches="tight")
/content/autonomous_driving/simulator.py:227: UserWarning: Glyph 24452 (\N{CJK UNIFIED IDEOGRAPH-5F84}) missing from font(s) DejaVu Sa
plt.savefig(save_path, dpi=150, bbox_inches="tight")
/content/autonomous_driving/simulator.py:227: UserWarning: Glyph 27178 (\N{CJK UNIFIED IDEOGRAPH-6A2A}) missing from font(s) DejaVu Sa
plt.savefig(save_path, dpi=150, bbox_inches="tight")
/content/autonomous_driving/simulator.py:227: UserWarning: Glyph 36716 (\N{CJK UNIFIED IDEOGRAPH-8F6C}) missing from font(s) DejaVu Sa
plt.savefig(save_path, dpi=150, bbox_inches="tight")
/content/autonomous_driving/simulator.py:227: UserWarning: Glyph 35282 (\N{CJK UNIFIED IDEOGRAPH-89D2}) missing from font(s) DejaVu Sa
plt.savefig(save_path, dpi=150, bbox_inches="tight")
/content/autonomous_driving/simulator.py:227: UserWarning: Glyph 36319 (\N{CJK UNIFIED IDEOGRAPH-8DDF}) missing from font(s) DejaVu Sa
plt.savefig(save_path, dpi=150, bbox_inches="tight")
/content/autonomous_driving/simulator.py:227: UserWarning: Glyph 36394 (\N{CJK UNIFIED IDEOGRAPH-8E2A}) missing from font(s) DejaVu Sa
plt.savefig(save_path, dpi=150, bbox_inches="tight")
/content/autonomous_driving/simulator.py:227: UserWarning: Glyph 26102 (\N{CJK UNIFIED IDEOGRAPH-65F6}) missing from font(s) DejaVu Sa
plt.savefig(save_path, dpi=150, bbox_inches="tight")
/content/autonomous_driving/simulator.py:227: UserWarning: Glyph 38388 (\N{CJK UNIFIED IDEOGRAPH-95F4}) missing from font(s) DejaVu Sa
plt.savefig(save_path, dpi=150, bbox_inches="tight")
/content/autonomous_driving/simulator.py:227: UserWarning: Glyph 27493 (\N{CJK UNIFIED IDEOGRAPH-6B65}) missing from font(s) DejaVu Sa
plt.savefig(save_path, dpi=150, bbox_inches="tight")
/content/autonomous_driving/simulator.py:227: UserWarning: Glyph 29702 (\N{CJK UNIFIED IDEOGRAPH-7406}) missing from font(s) DejaVu Sa
plt.savefig(save_path, dpi=150, bbox_inches="tight")
/content/autonomous_driving/simulator.py:227: UserWarning: Glyph 24819 (\N{CJK UNIFIED IDEOGRAPH-60F3}) missing from font(s) DejaVu Sa
plt.savefig(save_path, dpi=150, bbox_inches="tight")
/content/autonomous_driving/simulator.py:227: UserWarning: Glyph 39044 (\N{CJK UNIFIED IDEOGRAPH-9884}) missing from font(s) DejaVu Sa
plt.savefig(save_path, dpi=150, bbox_inches="tight")
/content/autonomous_driving/simulator.py:227: UserWarning: Glyph 27979 (\N{CJK UNIFIED IDEOGRAPH-6D4B}) missing from font(s) DejaVu Sa
plt.savefig(save_path, dpi=150, bbox_inches="tight")
/content/autonomous_driving/simulator.py:227: UserWarning: Glyph 35823 (\N{CJK UNIFIED IDEOGRAPH-8BEF}) missing from font(s) DejaVu Sa
plt.savefig(save_path, dpi=150, bbox_inches="tight")
/content/autonomous_driving/simulator.py:227: UserWarning: Glyph 24046 (\N{CJK UNIFIED IDEOGRAPH-5DEE}) missing from font(s) DejaVu Sa
plt.savefig(save_path, dpi=150, bbox_inches="tight")
/content/autonomous_driving/simulator.py:227: UserWarning: Glyph 20687 (\N{CJK UNIFIED IDEOGRAPH-50CF}) missing from font(s) DejaVu Sa
plt.savefig(save_path, dpi=150, bbox_inches="tight")
/content/autonomous_driving/simulator.py:227: UserWarning: Glyph 32032 (\N{CJK UNIFIED IDEOGRAPH-7D20}) missing from font(s) DejaVu Sa
plt.savefig(save_path, dpi=150, bbox_inches="tight")
/content/autonomous_driving/simulator.py:227: UserWarning: Glyph 36947 (\N{CJK UNIFIED IDEOGRAPH-9053}) missing from font(s) DejaVu Sa
plt.savefig(save_path, dpi=150, bbox_inches="tight")
/content/autonomous_driving/simulator.py:227: UserWarning: Glyph 20559 (\N{CJK UNIFIED IDEOGRAPH-504F}) missing from font(s) DejaVu Sa
plt.savefig(save_path, dpi=150, bbox_inches="tight")
/content/autonomous_driving/simulator.py:227: UserWarning: Glyph 31163 (\N{CJK UNIFIED IDEOGRAPH-79BB}) missing from font(s) DejaVu Sa
plt.savefig(save_path, dpi=150, bbox_inches="tight")
/content/autonomous_driving/simulator.py:227: UserWarning: Glyph 30417 (\N{CJK UNIFIED IDEOGRAPH-76D1}) missing from font(s) DejaVu Sa
plt.savefig(save_path, dpi=150, bbox_inches="tight")
/content/autonomous_driving/simulator.py:227: UserWarning: Glyph 36793 (\N{CJK UNIFIED IDEOGRAPH-8FB9}) missing from font(s) DejaVu Sa
plt.savefig(save_path, dpi=150, bbox_inches="tight")
/content/autonomous_driving/simulator.py:227: UserWarning: Glyph 30028 (\N{CJK UNIFIED IDEOGRAPH-754C}) missing from font(s) DejaVu Sa
plt.savefig(save_path, dpi=150, bbox_inches="tight")
模拟结果图已保存: /content/autonomous_driving/output/smallnet_20260710_135953_simulation.png
Figure(1200x1000)

开始借助 AI 编写或生成代码。

nvidia-smi									
Fri Jul 10 13:48:14 2026									
NVIDIA-SMI 580.82.07			Driver Version: 580.82.07			CUDA Version: 13.0			
GPU	Name		Persistence-M	Bus-Id	Disp. A	Volatile	Uncorr.	ECC	
Fan	Temp	Perf	Pwr:Usage/Cap		Memory-Usage	GPU-Util	Compute M.		
							MIG M.		
0	Tesla T4		Off		00000000:00:04.0 Off				0

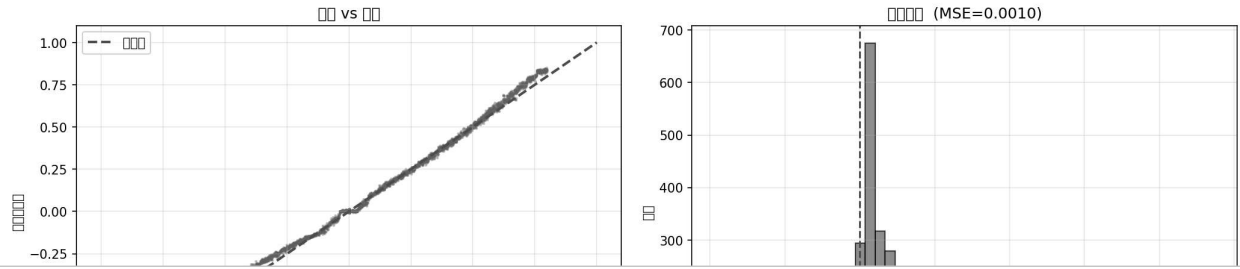
N/A	44C	P8	14W / 70W	0MiB / 15360MiB	0%	Default
						N/A
Processes:						
GPU	GI	CI	PID	Type	Process name	GPU Memory
	ID	ID				Usage
No running processes found						

```

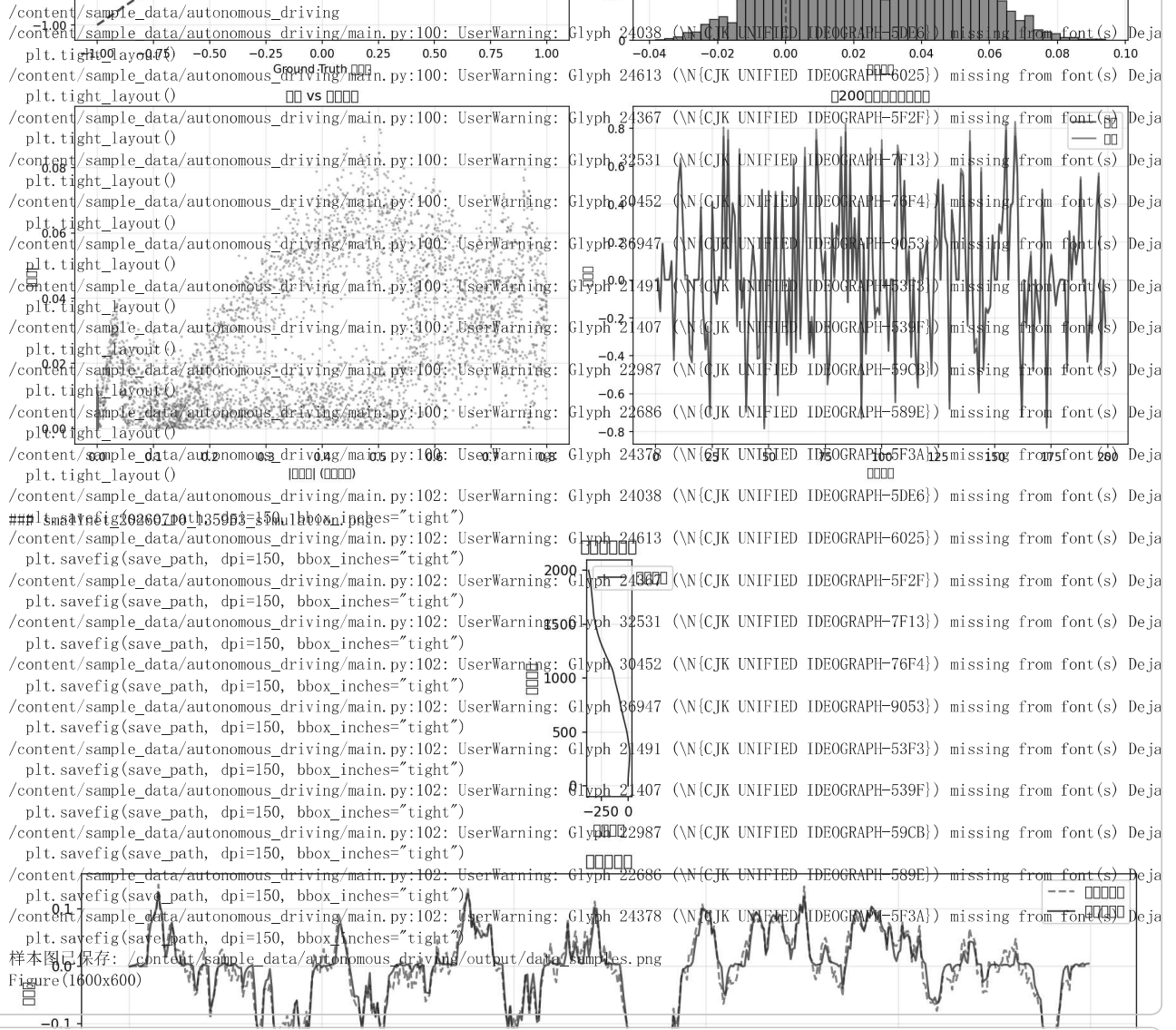
from IPython.display import Image, display
import os
output_dir = "output"
images = [f for f in os.listdir(output_dir) if f.endswith(".png")]
images.sort()
for img_name in images:
    print(f"\n### {img_name}")
    display(Image(filename=f"{output_dir}/{img_name}"))

```

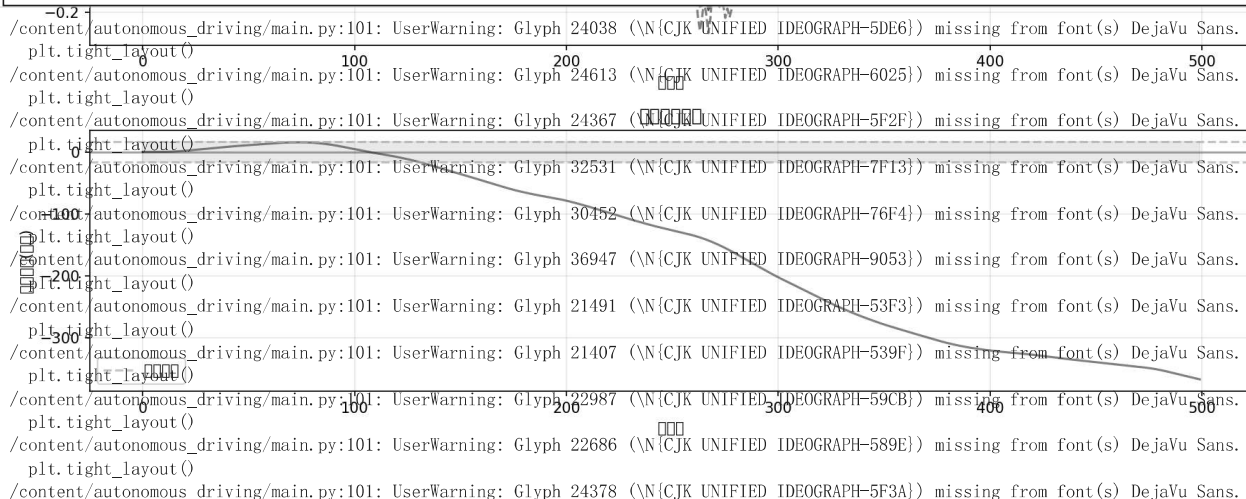

smallnet_20260710_135953_evaluation.png



```
%cd /content/sample_data/autonomous_driving
!python main.py demo
```



```
!python main.py demo
```



```
plt.tight_layout()
/content/autonomous_driving/main.py:103: UserWarning: Glyph 24038 (\N{CJK UNIFIED IDEOGRAPH-5DE6}) missing from font(s) DejaVu Sans.
plt.savefig(save_path, dpi=150, bbox_inches="tight")
/content/autonomous_driving/main.py:103: UserWarning: Glyph 24613 (\N{CJK UNIFIED IDEOGRAPH-6025}) missing from font(s) DejaVu Sans.
plt.savefig(save_path, dpi=150, bbox_inches="tight")
/content/autonomous_driving/main.py:103: UserWarning: Glyph 24367 (\N{CJK UNIFIED IDEOGRAPH-5F2F}) missing from font(s) DejaVu Sans.
plt.savefig(save_path, dpi=150, bbox_inches="tight")
/content/autonomous_driving/main.py:103: UserWarning: Glyph 32531 (\N{CJK UNIFIED IDEOGRAPH-7F13}) missing from font(s) DejaVu Sans.
plt.savefig(save_path, dpi=150, bbox_inches="tight")
/content/autonomous_driving/main.py:103: UserWarning: Glyph 30452 (\N{CJK UNIFIED IDEOGRAPH-76F4}) missing from font(s) DejaVu Sans.
plt.savefig(save_path, dpi=150, bbox_inches="tight")
/content/autonomous_driving/main.py:103: UserWarning: Glyph 36947 (\N{CJK UNIFIED IDEOGRAPH-9053}) missing from font(s) DejaVu Sans.
plt.savefig(save_path, dpi=150, bbox_inches="tight")
/content/autonomous_driving/main.py:103: UserWarning: Glyph 21491 (\N{CJK UNIFIED IDEOGRAPH-53F3}) missing from font(s) DejaVu Sans.
plt.savefig(save_path, dpi=150, bbox_inches="tight")
/content/autonomous_driving/main.py:103: UserWarning: Glyph 21407 (\N{CJK UNIFIED IDEOGRAPH-539F}) missing from font(s) DejaVu Sans.
plt.savefig(save_path, dpi=150, bbox_inches="tight")
/content/autonomous_driving/main.py:103: UserWarning: Glyph 22987 (\N{CJK UNIFIED IDEOGRAPH-59CB}) missing from font(s) DejaVu Sans.
plt.savefig(save_path, dpi=150, bbox_inches="tight")
/content/autonomous_driving/main.py:103: UserWarning: Glyph 22686 (\N{CJK UNIFIED IDEOGRAPH-589E}) missing from font(s) DejaVu Sans.
plt.savefig(save_path, dpi=150, bbox_inches="tight")
/content/autonomous_driving/main.py:103: UserWarning: Glyph 24378 (\N{CJK UNIFIED IDEOGRAPH-5F3A}) missing from font(s) DejaVu Sans.
plt.savefig(save_path, dpi=150, bbox_inches="tight")
样本图已保存: /content/autonomous_driving/output/data_samples.png
Figure (1600x600)
```

```
%cd /content/autonomous_driving
```

```
/content/autonomous_driving
```

```
!python main.py preview --weights models/checkpoints/smallnet_20260710_135953/best.weights.h5 --mode scene --frames 8
```

```
/content/autonomous_driving/preview.py:141: UserWarning: Glyph 20559 (\N{CJK UNIFIED IDEOGRAPH-504F}) missing from font(s) DejaVu Sans
plt.tight_layout()
/content/autonomous_driving/preview.py:141: UserWarning: Glyph 31163 (\N{CJK UNIFIED IDEOGRAPH-79BB}) missing from font(s) DejaVu Sans
plt.tight_layout()
/content/autonomous_driving/preview.py:143: UserWarning: Glyph 22330 (\N{CJK UNIFIED IDEOGRAPH-573A}) missing from font(s) DejaVu Sans
plt.savefig(save_path, dpi=150, bbox_inches="tight")
/content/autonomous_driving/preview.py:143: UserWarning: Glyph 26223 (\N{CJK UNIFIED IDEOGRAPH-666F}) missing from font(s) DejaVu Sans
plt.savefig(save_path, dpi=150, bbox_inches="tight")
/content/autonomous_driving/preview.py:143: UserWarning: Glyph 26354 (\N{CJK UNIFIED IDEOGRAPH-66F2}) missing from font(s) DejaVu Sans
plt.savefig(save_path, dpi=150, bbox_inches="tight")
/content/autonomous_driving/preview.py:143: UserWarning: Glyph 29575 (\N{CJK UNIFIED IDEOGRAPH-7387}) missing from font(s) DejaVu Sans
plt.savefig(save_path, dpi=150, bbox_inches="tight")
/content/autonomous_driving/preview.py:143: UserWarning: Glyph 27169 (\N{CJK UNIFIED IDEOGRAPH-6A21}) missing from font(s) DejaVu Sans
plt.savefig(save_path, dpi=150, bbox_inches="tight")
/content/autonomous_driving/preview.py:143: UserWarning: Glyph 22411 (\N{CJK UNIFIED IDEOGRAPH-578B}) missing from font(s) DejaVu Sans
plt.savefig(save_path, dpi=150, bbox_inches="tight")
/content/autonomous_driving/preview.py:143: UserWarning: Glyph 20915 (\N{CJK UNIFIED IDEOGRAPH-51B3}) missing from font(s) DejaVu Sans
plt.savefig(save_path, dpi=150, bbox_inches="tight")
/content/autonomous_driving/preview.py:143: UserWarning: Glyph 31574 (\N{CJK UNIFIED IDEOGRAPH-7B56}) missing from font(s) DejaVu Sans
plt.savefig(save_path, dpi=150, bbox_inches="tight")
/content/autonomous_driving/preview.py:143: UserWarning: Glyph 29702 (\N{CJK UNIFIED IDEOGRAPH-7406}) missing from font(s) DejaVu Sans
plt.savefig(save_path, dpi=150, bbox_inches="tight")
```